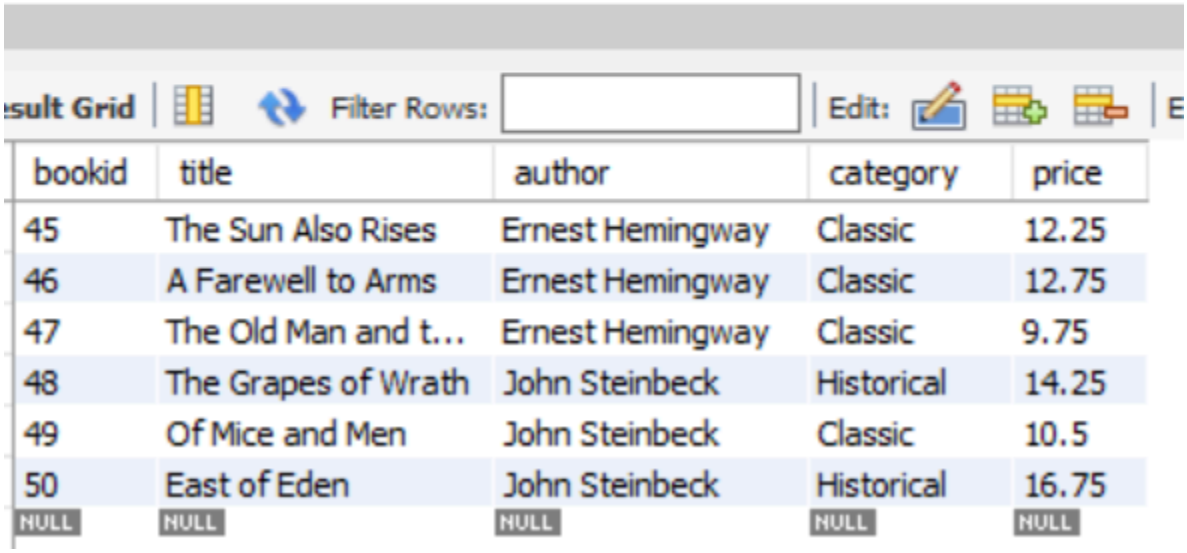
Title: Database Assignment 5
Your Name: Jack Saunders
Date: 11/5/2026

SQL setup:

```
1  create database HW5new;
2  use HW5new;
3
4  create table books (
5      bookid integer primary key,
6      title varchar(45),
7      author varchar(45),
8      category varchar(20),
9      price double
10 );
11
12 select * from books;
```



bookid	title	author	category	price
45	The Sun Also Rises	Ernest Hemingway	Classic	12.25
46	A Farewell to Arms	Ernest Hemingway	Classic	12.75
47	The Old Man and t...	Ernest Hemingway	Classic	9.75
48	The Grapes of Wrath	John Steinbeck	Historical	14.25
49	Of Mice and Men	John Steinbeck	Classic	10.5
50	East of Eden	John Steinbeck	Historical	16.75
NULL	NULL	NULL	NULL	NULL

Node server.js code:

```
// set up an HTTP server off port 3000
const express = require("express");
const app = express();
const port = 3000;

// set up mysql connections using the modern 'mysql2/promise' driver
const mysql = require('mysql2/promise');

// import the pool
const pool = require('./mysql');

// receiving user input using /get route and async function
app.get("/", async (req, res) => {
   const format = req.query.format || "json";

// columns the user is allowed to search by (must match your table's column
names)
   const SEARCHABLE = ["book_id", "title", "author", "category", "price"];

// build the WHERE clause from whichever parameters were supplied
   const conditions = [];
   const params = [];
   for (const column of SEARCHABLE)
       if (req.query[column]) {
           conditions.push(`${column} = ?`); // column name comes from the
whitelist
           params.push(req.query[column]);   // value is passed as a parameter
       }

// input validation: at least one search field is required
   if (conditions.length === 0)
       return res.status(400).send("Error: Please provide a search parameter in
the URL, e.g. ?author=George Orwell");

   const sqlQuery = "SELECT * FROM HW5new.books WHERE " + conditions.join(" AND
");
   try {
       // execute the query using pool.execute()
       const [results, fields] = await pool.execute(sqlQuery, params);
       console.log(`Executed query: ${sqlQuery} with category ${params}`);

       // check for formatting request
       if (format == "text"){
           let html = `<h1>Search Results</h1>`;
           results.forEach(book => {
               html += `
               <div>
                   <h2>Title: ${book.title}</h2>
```

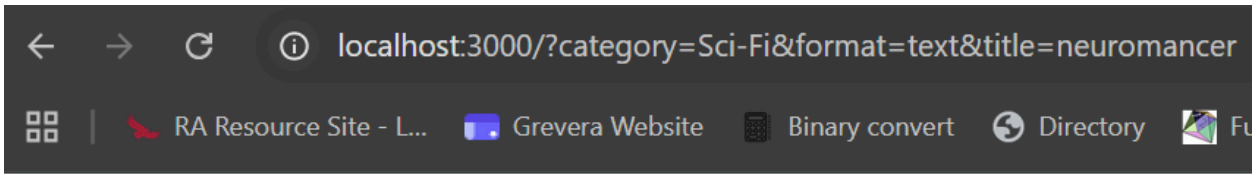
```
                    <p>Book ID: ${book.bookid}</p>
                    <p>Author: ${book.author}</p>
                    <p>Category: ${book.category}</p>
                    <p>Price: $ ${book.price}</p>
                </div>
                <hr>
                `;
            });
            // send the results back to the browser
            res.status(200).type('html').send(html);
        }else res.status(200).json(results);

    }catch (err) {
        // log the detailed error on the server
        console.error("Database error:", err.message);
        // Send a generic, non-revealing error to the client for security
        res.status(500).send("A database error occurred.");}
});

// receiving user input for format of results
app.get("/format", async (req,res) => {
    const format = req.query.format;
    if (format == "text"){
        let html = `<h1>Books in Database with Category: ${category}</h1>`;
        results.forEach(book => {
            html += `
            <div>
                <h2>Title: ${book.title}</h2>
                <p>Book ID: ${book.bookid}</p>
                <p>Author: ${book.author}</p>
                <p>Category: ${book.category}</p>
                <p>Price: $ ${book.price}</p>
            </div>
            <hr>
            `;
        });
        res.type('html').send(html);
    } else res.json(results);
})

// wait indefinitely in a loop
app.listen(port, function () {
    console.log(`Server running on http://localhost:${port}`);
});
```

Examples:



Search Results

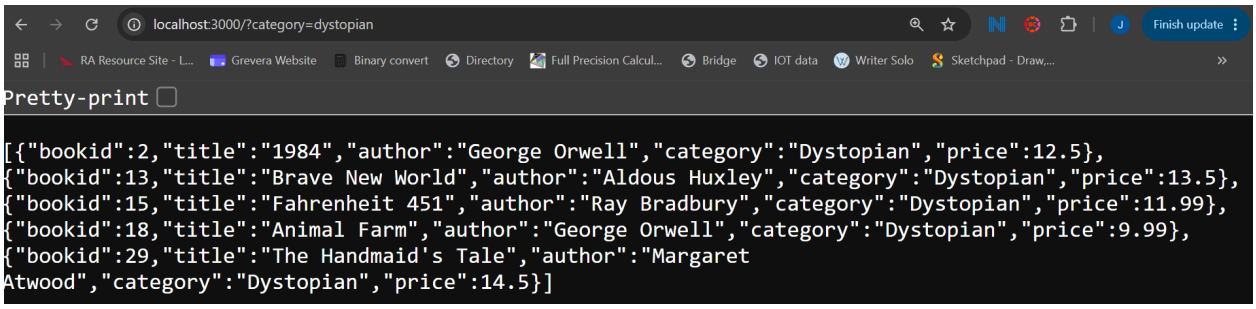
Title: Neuromancer

Book ID: 30

Author: William Gibson

Category: Sci-Fi

Price: $ 15




```
[{"bookid":2,"title":"1984","author":"George Orwell","category":"Dystopian","price":12.5},
{"bookid":13,"title":"Brave New World","author":"Aldous Huxley","category":"Dystopian","price":13.5},
{"bookid":15,"title":"Fahrenheit 451","author":"Ray Bradbury","category":"Dystopian","price":11.99},
{"bookid":18,"title":"Animal Farm","author":"George Orwell","category":"Dystopian","price":9.99},
{"bookid":29,"title":"The Handmaid's Tale","author":"Margaret
Atwood","category":"Dystopian","price":14.5}]
```